CES production – beyond Cobb-Douglas

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Purpose: The goal of this note is to re-do our calculations of the price of value added and the supply/demand influence calculations when we have CES production instead of Cobb-Douglas.

Problem: With Cobb-Douglas the output prices are log-linear in input prices and with Leontief the output prices are linear in input prices. With CES more generally, neither are true and value added prices become a non-linear function of output prices. This makes writing a decomposition much trickier. Let's see how far we can go.

Calculating the price of value added and Sales

Consumption remains Cobb-Douglas in products:

$$C_t = \prod_p C_{i,t}^{\omega_{PCE,i,t}}$$

where $\omega_{PCE,i,t}$ are the expenditure weights from the BEA over all $\mathcal{P}$ products with individual products denoted by i. Note that if I highlight something in green then we have data on it.

Production is now given by:

$$Y_{i,t} = \left[(1 - c_i) VA_{i,t}^{\frac{\epsilon-1}{\epsilon}} + c_i \sum_j \varpi_{j \to i} M_{j \to i,t}^{\frac{\epsilon-1}{\epsilon}} \right]^{\frac{\epsilon}{\epsilon-1}}$$

where $M_{i,t}$ is intermediate inputs used by i and $M_{j \to i,t}$ are intermediates of product j used by i in production.

Assuming firms optimize profits taking prices as given:

$$\max_{Y_{i,t}, VA_{i,t}, \{M_{j \to i,t}\}_j} P_{i,t} Y_{i,t} - P_{VA,i,t} VA_{i,t} - \sum_j P_{j,t} M_{j \to i,t}$$

Standard optimization gives us the following formula for output prices:

$$P_{i,t} = \left[(1-c_i)^{\epsilon} P_{VA,i,t}^{1-\epsilon} + c_i^{\epsilon} \sum_j \omega_{j\to i}^{\epsilon} {P_{j,t}}^{1-\epsilon} \right]^{\frac{1}{1-\epsilon}} \tag{1}$$

and the following formula for the expenditure shares that each sector i spends on inputs from j:

$$\omega_{j\to i,t} \equiv \frac{P_{j,t} M_{j\to i,t}}{P_{i,t} Y_{i,t}} = (c_i \omega_{j\to i})^{\epsilon} \left(\frac{P_{j,t}}{P_{i,t}} \right)^{1-\epsilon}$$

with the share of expenditure on value added given by: $\omega_{VA,i,t} \equiv 1 - \sum_j \omega_{j\to i,t}$.

Note that because we only have input-output data in 2017, we are assuming that the expenditure shares for inputs never change:

$$\omega_{j\to i,t} = \omega_{j\to i,2017}$$

Unlike the Cobb-Douglas case, Equation 1 cannot be linearly inverted! However, note that the expression is linear if instead of solving for $P_{i,t}$ in terms of $P_{VA,i,t}$ and $P_{j,t}$, we solve for $P_{i,t}^{1-\epsilon}$ and $P_{VA,i,t}^{1-\epsilon}$.

Let's also try to solve for the (gross) change in output prices: $\hat{\Pi}_{i,t}^{1-\epsilon} \equiv P_{i,t}/P_{i,t-1}$ (where in general $\hat{X}_t \equiv X_t/X_{t-1}$). Given that the parameters are assumed to be constant we can write this as:

$$\begin{aligned}
\hat{\Pi}_{i,t}^{1-\epsilon} &= \left(\frac{P_{i,t}}{P_{i,t-1}} \right)^{1-\epsilon} = \frac{(1-c_i)^{\epsilon} P_{VA,i,t}^{1-\epsilon} + c_i^{\epsilon} \sum_j \omega_{j\to i}^{\epsilon} {P_{j,t}}^{1-\epsilon}}{(1-c_i)^{\epsilon} P_{VA,i,t-1}^{1-\epsilon} + c_i^{\epsilon} \sum_j \omega_{j\to i}^{\epsilon} {P_{j,t-1}}^{1-\epsilon}} \\
&= \frac{(1-c_i)^{\epsilon} \hat{\Pi}_{VA,i,t}^{1-\epsilon} P_{VA,i,t-1}^{1-\epsilon} + c_i^{\epsilon} \sum_j \omega_{j\to i}^{\epsilon} \hat{\Pi}_{j,t}^{1-\epsilon} {P_{j,t-1}}^{1-\epsilon}}{(1-c_i)^{\epsilon} P_{VA,i,t-1}^{1-\epsilon} + c_i^{\epsilon} \sum_j \omega_{j\to i}^{\epsilon} {P_{j,t-1}}^{1-\epsilon}} \\
&= \frac{(1-c_i)^{\epsilon} P_{VA,i,t-1}^{1-\epsilon}}{(1-c_i)^{\epsilon} P_{VA,i,t-1}^{1-\epsilon} + c_i^{\epsilon} \sum_j \omega_{j\to i}^{\epsilon} {P_{j,t-1}}^{1-\epsilon}} \hat{\Pi}_{VA,i,t}^{1-\epsilon} + \sum_j \frac{c_i^{\epsilon} \omega_{j\to i}^{\epsilon} {P_{j,t-1}}^{1-\epsilon}}{(1-c_i)^{\epsilon} P_{VA,i,t-1}^{1-\epsilon} + c_i^{\epsilon} \sum_j \omega_{j\to i}^{\epsilon} {P_{j,t-1}}^{1-\epsilon}} \hat{\Pi}_{j,t}^{1-\epsilon} \\
&= \underbrace{\frac{(1-c_i)^{\epsilon} P_{VA,i,t-1}^{1-\epsilon}}{{P_{i,t-1}}^{1-\epsilon}}}_{\equiv \omega_{VA,i,t-1}} \hat{\Pi}_{VA,i,t}^{1-\epsilon} + \sum_j \underbrace{\frac{c_i^{\epsilon} \omega_{j\to i}^{1-\epsilon} {P_{j,t-1}}^{1-\epsilon}}{{P_{i,t-1}}^{1-\epsilon}}}_{\equiv \omega_{j\to i,t-1}} \hat{\Pi}_{j,t}^{1-\epsilon} \\
&= \omega_{VA,i,t-1} \hat{\Pi}_{VA,i,t}^{1-\epsilon} + \sum_j \omega_{j\to i,t-1} \hat{\Pi}_{j,t}^{1-\epsilon} \\
&= \omega_{VA,i,2017} \hat{\Pi}_{VA,i,t}^{1-\epsilon} + \sum_j \omega_{j\to i,2017} \hat{\Pi}_{j,t}^{1-\epsilon}
\end{aligned} \tag{2}$$

where the last line follows by the assumption that expenditure shares on products never change.

And thus output price inflation (raised to the power of $1-\epsilon$) is linearly related to value added inflation and output prices in other sectors (both raised to the power of $1-\epsilon$).

Let's define the following matrix $\boldsymbol{A} \equiv \{\omega_{j\to i,2017}\}_{j,i}$ carefully. There is an issue becuase Ben and Nick like to define our input-output matrices differently ($\boldsymbol{A}_{Ben} = \boldsymbol{A}'_{Nick}$). This is leading to

math errors that affects Connor!. I will fix this by adopting Ben's convention whenever I use the matrix $\boldsymbol{A}$ and Nick's convention whenever we use the matrix $\boldsymbol{\Omega}$. Apologies Connor!

Ok, let's return to defining $\boldsymbol{A}$ and the associated math carefully. Note that Ben's definition of $\boldsymbol{A}$ has the *columns* sum to the share of sales spent on intermediates. *This should be the definitive test of whether $\boldsymbol{A}$ is the correct way around!*

To be clear:

$$\boldsymbol{A} = \begin{bmatrix} a_{11} & a_{12} & & a_{1\mathcal{P}} \\ a_{21} & a_{22} & \cdots & a_{2\mathcal{P}} \\ \vdots & \vdots & & \vdots \\ \underbrace{a_{\mathcal{P}1}}_{\sum_j a_{j,1}<1} & \underbrace{a_{\mathcal{P}2}}_{\sum_j a_{j,2}<1} & & \underbrace{a_{\mathcal{P}\mathcal{P}}}_{\sum_j a_{j,\mathcal{P}}<1} \end{bmatrix}$$

and let $\underline{X}_t$ be the vector of $X_{i,t}$. Finally let $\mathbb{1}_{\mathcal{P}}$ be a $\mathcal{P}$ x 1 vector of ones (where $\mathcal{P}$ is the number of products).

Another test that $\boldsymbol{A}$ is the correct orientation is to calculate $\mathbb{1}'_{\mathcal{P}}\boldsymbol{A}$ and confirm all elements are less than 1.

With these matrix definitions, Equation 2 becomes:

$$\begin{aligned} \underline{\hat{\Pi}}_t^{1-\epsilon} &= diag(1 - \boldsymbol{A}'\mathbb{1}_{\mathcal{P}})\underline{\hat{\Pi}}_{VA,t}^{1-\epsilon} + \boldsymbol{A}'\underline{\hat{\Pi}}_t^{-1\epsilon} \\ \Leftrightarrow \underline{\hat{\Pi}}_{VA,t}^{1-\epsilon} &= [diag(1 - \boldsymbol{A}'\mathbb{1}_{\mathcal{P}}]^{-1}(I - \boldsymbol{A}')\underline{\hat{\Pi}}_t^{1-\epsilon} \end{aligned} \tag{3}$$

Next let's calculate Sales. We have the following market clearing condition for each good:

$$Y_{i,t} = C_{i,t} + \sum_j M_{i\to j,t}$$

Since we assume the same price for all customers, let's multiply by the price $P_{i,t}$ and use the definitions of the expenditure shares to get the following expressions:

$$\begin{aligned} Sales_{i,t} \equiv P_{i,t}Y_{i,t} &= \underbrace{P_{i,t}C_{i,t}}_{\text{BEA expenditure data}} + \sum_j P_{j,t}M_{i\to j,t} \\ &= P_{i,t}C_{i,t} + \sum_j \omega_{i\to j,2017} Sales_{j,t} \end{aligned}$$

In matrix notation we have:

$$\begin{aligned} \underline{Sales}_t &= \underline{PC}_{i,t} + \underbrace{\boldsymbol{A}}_{\text{no transpose}} \underline{Sales}_t \\ \Leftrightarrow \underline{Sales}_t &= (1 - \boldsymbol{A})^{-1}\underline{PC}_{i,t} \end{aligned}$$

Inflation the way Shapiro does it

Note that Shapiro calculates CPI inflation as:

$$\pi_{CPI,t} \equiv \sum_i \omega_{PCE,i,t} \left(\frac{P_{i,t}}{P_{i,t-1}} - 1 \right) = \underline{\omega_{PCE,t}}' \underline{\pi_t}$$

with $\mathbb{1}'_{\mathcal{P}} \omega_{PCE,t} = 1$.

Note that we have a problem here: we have a linear decomposition of $\underline{\Pi}_t^{1-\epsilon} = (\underline{\pi}_t + 1)^{1-\epsilon}$! Obviously a linear decomposition of $\underline{\Pi}_t^{1-\epsilon}$ is not a linear decomposition of $\underline{\pi}_t$!

But we can cheat a bit here. Define $S_{i,t} \equiv (\hat{\Pi}_{i,t} - 1)/(\hat{\Pi}_{i,t}^{1-\epsilon} - 1)$. We have:

$$\begin{aligned} \pi_t &= diag(\underline{S}_t) \hat{\Pi}_t^{1-\epsilon} \\ &= diag(\underline{S}_t)(I - \boldsymbol{A}')^{-1} diag(1 - \boldsymbol{A}' \mathbb{1}_{\mathcal{P}})(\hat{\underline{\Pi}}_{VA,t}^{1-\epsilon} - 1) \end{aligned}$$

Given this, we can then decompose Shaprio CPI into:

$$\begin{aligned} \pi_{CPI,t} &= \omega_{PCE,t}' diag(\underline{S}_t)(\hat{\Pi}_t^{1-\epsilon} - 1) \\ &= \omega_{PCE,t}' diag(\underline{S}_t)(I - \boldsymbol{A}')^{-1} diag(1 - \boldsymbol{A}' \mathbb{1}_{\mathcal{P}})(\hat{\underline{\Pi}}_{VA,t}^{1-\epsilon} - 1) \end{aligned}$$

Why is this cheating? Because $S_{i,t}$ is clearly affected by $\hat{\underline{\Pi}}_{VA,t}$ but we are pretending it is not. Effectively we are saying that if (elasticity scaled) value added cost changes in sector j contributes 30% to the elasticity-scaled price change in sector i — $\hat{\Pi}_{i,t}^{1-\epsilon}$ — then it also contributes 30% to the unscaled price change in sector i! Obviously this is mathematically incorrect: the contribution of a factor to X is in general not equivalent to its contribution to $X^{1-\epsilon}$. But I think this is ok as a fudge for the current exercise. Since we do have a linear decomposition in the elasticity-scaled price changes, I don't think it matters too much to say this is equivalent to having the same share be allocated to the non-scaled price change.

What to do when the product set changes?

Implicitly we are approximating $\omega_{j \to i,t-1}$ with $\omega_{j \to i,2017}$ so we need to use $t-1$ as the trigger for when products exist or not.

There are two cases:

1. **When a product enters in t.** It should <u>not</u> be in $\omega_{j \to i,t-1}$. Also, value added price changes for products that are just entering should be 1: $\hat{\Pi}_{VA,i,t} = 1$ for all i entering. To remove i from $\omega_{j \to i,t-1}$ let's continue to make the ith row and column zeros with a 1 on the ith diagonal:

$$\omega_{j \to i,t-1} = \begin{cases} \omega_{j \to i,2017} & \text{if } P_{i,t},\ P_{i,t-1},\ P_{j,t} \text{ and } P_{j,t-1} \text{exist} \\ 0 & \text{If } P_{i,t-1} \text{ or } P_{j,t-1} \text{does not exist} \end{cases}$$

   We also have:

$$\hat{\Pi}_{VA,i,t} = 1$$

   whenever i does not exist in $t-1$.

2. **When a product exits in t:** Because here we are using the lagged weights to calculate the change in prices, we should give the series non-zero weight. That means we calculate the change in value-added prices that are consistent with the product existing in t even though it doesn't.

Implementation

Here are my suggested steps to follow:

1. Check my math above. Any issues or questions email/call me.
2. Define
$$\omega_{j\to i,t-1} = \begin{cases} \omega_{j\to i,2017} & \text{if } P_{i,t},\ P_{i,t-1},\ P_{j,t} \text{ and } P_{j,t-1} \text{exist} \\ 0 & \text{If } P_{i,t-1} \text{ or } P_{j,t-1} \text{does not exist} \end{cases}$$
where note that when all products are observed this should be the same as the $\boldsymbol{A}$ matrix in Leontief. Quick check on this (sums across a row):
   - $\max\{\boldsymbol{A}'\mathbb{1}_{\mathcal{P}}\} < 1$
   - $\max\{\boldsymbol{A}'\mathbb{1}_{\mathcal{P}}\} > 0$

   If this does not hold, check $\boldsymbol{A}'$. It should hold for the transpose and then please proceed with the transpose of the input matrix $\boldsymbol{A}'$. If this does hold, continue without transposing.
3. Create time varying I-O matrices: $\boldsymbol{A}_t \equiv \{\omega_{j\to i,t}\}_{i,j}$ from step 1.
4. Calculate Sales according to:
$$\underline{Sales}_t = \left(1 - \underbrace{\boldsymbol{A}_t}_{\text{no transpose}}\right)^{-1} \underline{PC}_{i,t}$$
5. [Should trivially pass:] Check that every sales is always above 0. If it is ever negative try transposing $\boldsymbol{A}_t$. If that fails talk with Nick. Works with $\boldsymbol{A}$ or $\boldsymbol{A}'$. Only fails if negatives in the matrix $\boldsymbol{A}$ or if a column sum is larger than 1.
6. Pick an elasticity ϵ. To start with, let's do something like 0.3.
7. Construct output price inflation raised to the power of $1-\epsilon$: $\hat{\underline{\Pi}}_t^{1-\epsilon} \equiv \left(\frac{\underline{P}_t}{\underline{P}_{t-1}}\right)^{1-\epsilon}$.
8. Calculate $\boldsymbol{B}_t \equiv (I - \boldsymbol{A}_t')^{-1} diag(1 - \boldsymbol{A}_t'\mathbb{1}_{\mathcal{P}})$.
9. Confirm that $\boldsymbol{B}_t\mathbb{1}_{\mathcal{P}} = 1$ for all time periods t. Again, this will pass if $\boldsymbol{A}$ or $\boldsymbol{A}'$ are used. It is a math check not a test of $\boldsymbol{A}$ vs. $\boldsymbol{A}'$. It is a test of whether $\boldsymbol{A}$ has correct entries (column sums less than 1 and non-negative elements) and whether Nick has written the math correctly (or Connor has coded it correctly).

10. Calculate value added inflation consistent with this elasticity:

$$\underline{\hat{\Pi}}_{VA,t}^{1-\epsilon} = \boldsymbol{B}_t^{-1}\underline{\hat{\Pi}}_t^{1-\epsilon}$$

11. Confirm that $\underline{\hat{\Pi}}_t^{1-\epsilon} - \boldsymbol{B}_t\underline{\hat{\Pi}}_{VA,t}^{1-\epsilon} = \underline{0}$. If not, something is wrong and you need to check for errors.

12. Once we have value added inflation calculate net value added inflation:

$$\pi_{VA,i,t} = \left(\Pi_{VA,i,t}^{1-\epsilon}\right)^{\frac{1}{1-\epsilon}} - 1$$

13. Use sales and net value added inflation as LHS variables in the VAR analysis to get $D_{i,t}^{CES}$ to indicate if product i in time t is mostly demand or not.

14. Calculate $S_t \equiv \frac{\underline{\hat{\Pi}}_t - 1}{\underline{\hat{\Pi}}_t^{1-\epsilon} - 1}$.

15. Calculate $\pi_{i,t} \equiv \frac{P_{i,t}}{P_{i,t-1}}$ and confirm:

$$\pi_{i,t} = S_{i,t}\left(\hat{\Pi}_{i,t}^{1-\epsilon} - 1\right)$$

16. Calculate $\pi_{CPI,t} \equiv \sum_i \omega_{PCE,i,t}\pi_{i,t}$ and then confirm that:

$$\pi_{CPI,t} = \underline{\omega}_{PCE,t}'\underline{\pi}_t$$

If not, you made a coding error somewhere.

17. Calculate the weights:

$$\underline{\omega}_{VA,t} \equiv \underline{\omega}_{PCE,t}'(I - \boldsymbol{A}_t')^{-1}diag(I - \boldsymbol{A}_t'\mathbb{1}_{\mathcal{P}})$$

Note that $\omega_{VA,i,t}$ may not sum to 1 over i however, we do want $\omega_{VA,i,t} > 0$. You made an error if this is not the case (or Nick's intuition about this being above 0 is wrong).

18. Check that

$$\pi_{CPI,t} = \underline{\omega}_{VA,t}'\left(\underline{\hat{\Pi}}_{VA,t}^{1-\epsilon} - 1\right)$$

If not, you made an error or Nick made a math error (likely it was Nick).

19. Calculate the following:

$$\pi^{\text{Shapiro scale}}_{demand,t} = \sum_j D^{CES}_{i,t} \omega_{PCE,j,t} \pi_{j,t}$$

$$\pi^{\text{Shapiro scale}}_{supply,t} = \sum_j (1 - D^{CES}_{i,t}) \omega_{PCE,j,t} \pi_{j,t}$$

$$\pi^{\text{Shapiro shocks}}_{demand,t} = \sum_j D^{\text{Shapiro original}}_{i,t} \underline{\omega}_{VA,j,t} (\hat{\Pi}^{1-\epsilon}_{VA,j,t} - 1)$$

$$\pi^{\text{Shapiro shocks}}_{supply,t} = \sum_j (1 - D^{\text{Shapiro original}}_{i,t}) \underline{\omega}_{VA,j,t} (\hat{\Pi}^{1-\epsilon}_{VA,j,t} - 1)$$

$$\pi^{CES}_{demand,t} = \sum_j D^{CES}_{i,t} \underline{\omega}_{VA,j,t} (\hat{\Pi}^{1-\epsilon}_{VA,j,t} - 1)$$

$$\pi^{CES}_{supply,t} = \sum_j (1 - D^{CES}_{i,t}) \underline{\omega}_{VA,j,t} (\hat{\Pi}^{1-\epsilon}_{VA,j,t} - 1)$$

20. Do the Shapiro 12 month lag thing on all series (including $\pi_{CPI,t}$).

21. Produce 3 plots with:

    (a) $\pi^{12}_{CPI,t}$ as a line against bars of $\pi^{\text{Shapiro scale}}_{demand,t}$ and $\pi^{\text{Shapiro scale}}_{supply,t}$

    (b) $\pi^{12}_{CPI,t}$ as a line against bars of $\pi^{\text{Shapiro shocks}}_{demand,t}$ and $\pi^{\text{Shapiro shocks}}_{supply,t}$

    (c) $\pi^{12}_{CPI,t}$ as a line against bars of $\pi^{CES}_{demand,t}$ and $\pi^{CES}_{supply,t}$

22. If this all looks good, maybe try plot (c) with different ϵ? Maybe $\epsilon \in \{0.1, 0.6\}$ as well as 0.3.